Precipitation Reactions

A precipitation reaction occurs when two salt solutions are mixed and an insoluble solid (or precipitate) is formed.

Scientists have developed a set of rules for determining which salts are soluble in water and which are insoluble in water. These rules are in the tables below.

Soluble in Water

Soluble	**Exceptions**	
	Insoluble	**Slightly soluble**
Most chlorides	AgCl,	$PbCl_2$
Most bromides	AgBr,	$PbBr_2$
Most iodides	AgI, PbI_2	
All nitrates	No exceptions	
Most sulfates	$SrSO_4$, $BaSO_4$, $PbSO_4$	$CaSO_4$, Ag_2SO_4

Insoluble in Water

Insoluble	**Exceptions**	
	Soluble	**Slightly soluble**
Most hydroxides	NaOH, KOH, $Ba(OH)_2$	$Ca(OH)_2$, $Sr(OH)_2$
Most carbonates	Na_2CO_3, K_2CO_3, $(NH_4)_2CO_3$	
Most phosphates	Na_3PO_4, K_3PO_4, $(NH_4)_3PO_4$	

Use the two above tables to predict which of the salts below are soluble in water and which are insoluble.

1. $Al(OH)_3$
2. Na_2CO_3
3. $Ca_3(PO_4)_2$
4. NaCl
5. $AgNO_3$
6. CaI_2
7. $PbBr_2$
8. $CuSO_4$
9. $CaCO_3$
10. $NiCl_2$
11. $SnSO_4$
12. $CoCl_2$
13. AgBr
14. $ZnCO_3$

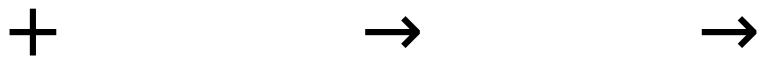
15. $Fe(OH)_2$
16. $Mg(NO_3)_2$
17. K_2SO_4
18. $BaSO_4$
19. Ag_2CO_3
20. $MgCl_2$

When salts are dissolved in water, they break up into separate ions. When two solutions are mixed, sometimes the attraction between two ions is greater than the attraction between the water molecules and the ions. When this happens, a solid will precipitate out of the solution.

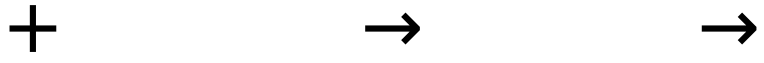


Silver Nitrate + Sodium chloride → Silver chloride + Sodium nitrate

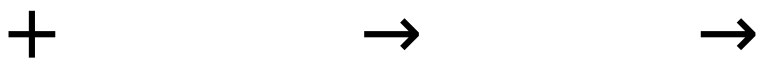
Predict what solid will form in each of the reactions below. Write what ions are contained in each beaker, then figure out what combinations of ions could form.



Lead (II) nitrate + Potassium iodide →



Calcium nitrate + Sodium carbonate →



Sodium hydroxide + Aluminium chloride